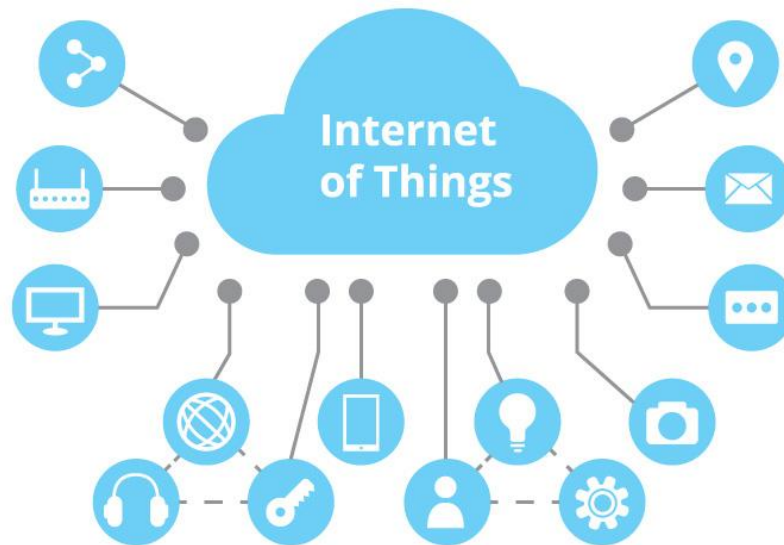


Internet of Things (IoT)



Chapter 3: Sensors and Actuators

Nguyen Tan No

Contents

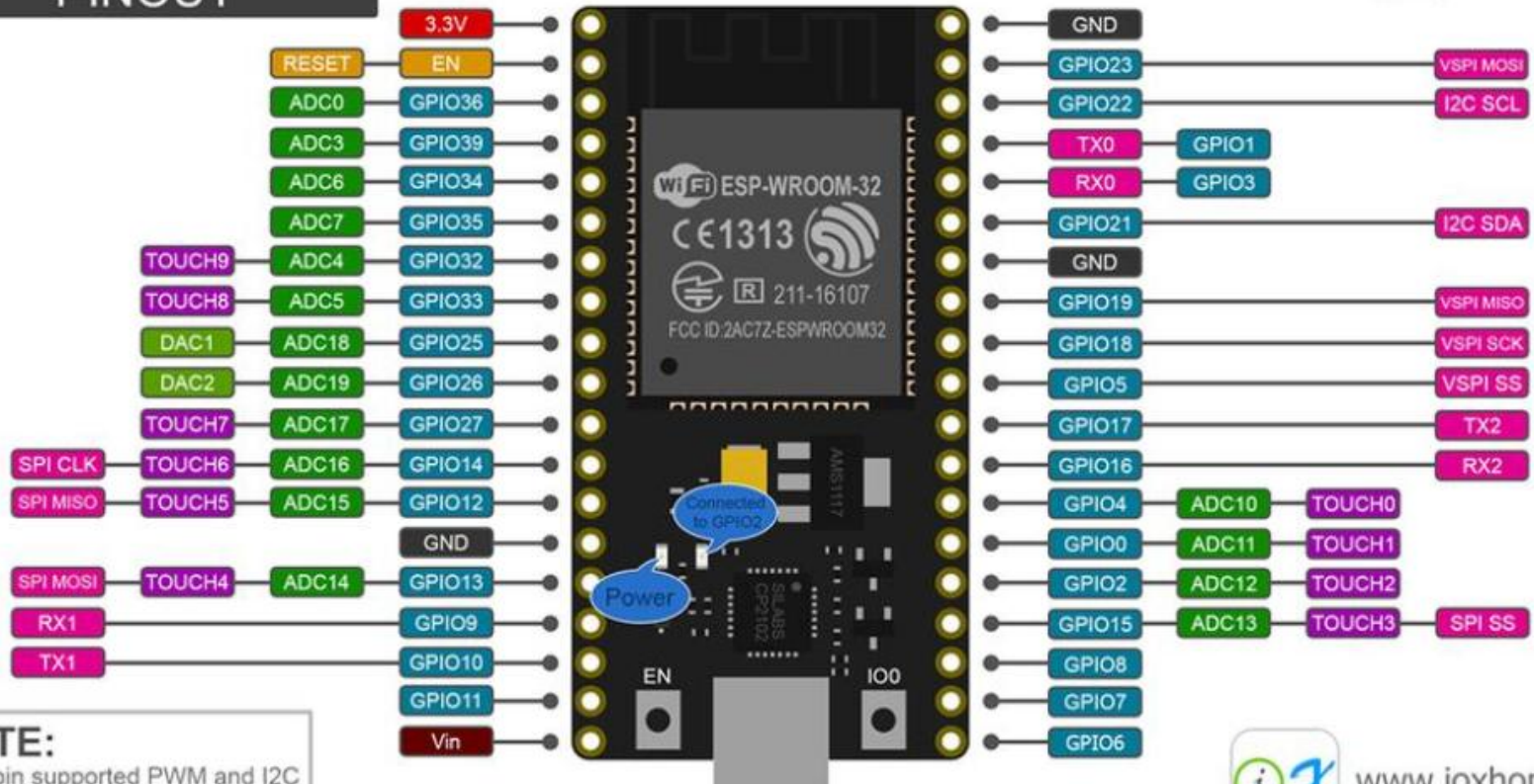
1. Sensors

2. Actuators

Contents

NodeMCU-32S

PINOUT



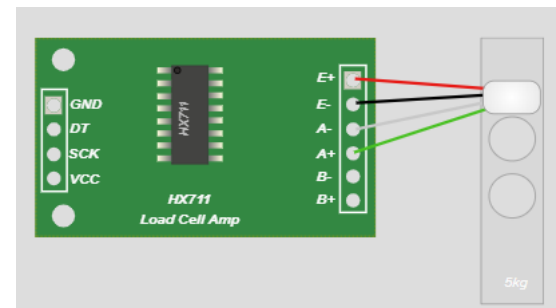
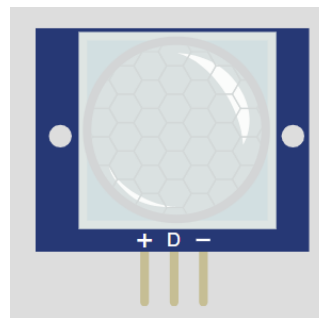
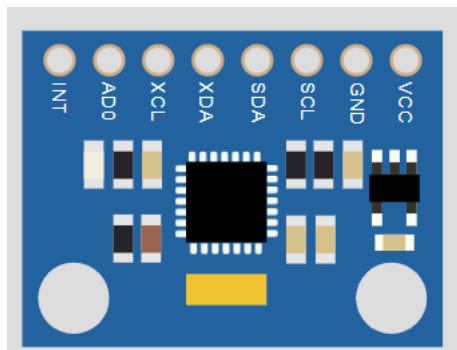
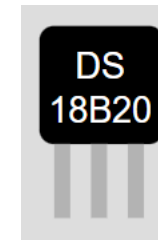
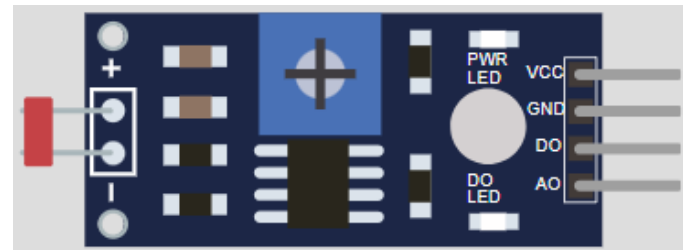
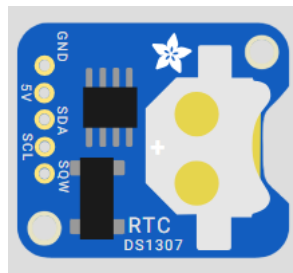
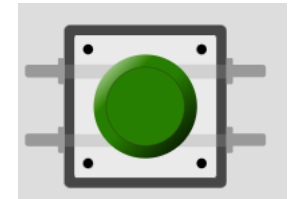
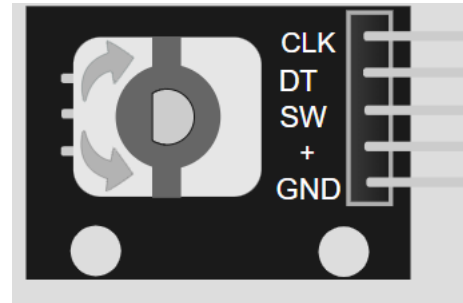
NOTE:

All pin supported PWM and I2C
Pin current 6mA (Max. 12mA)

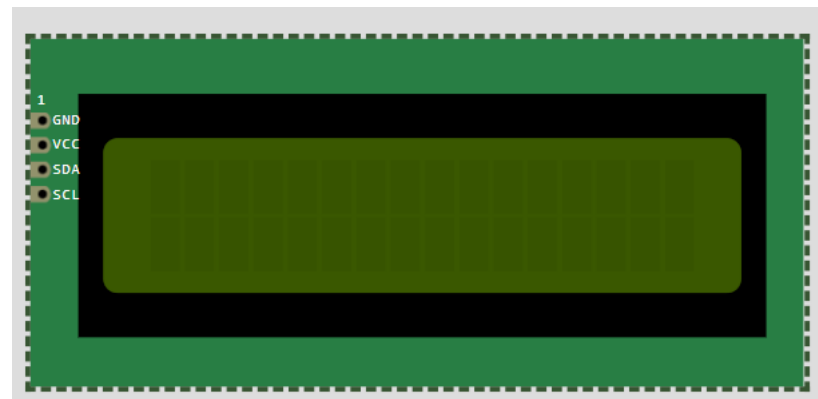
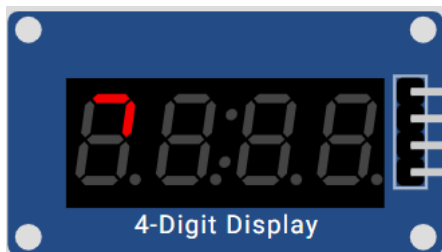
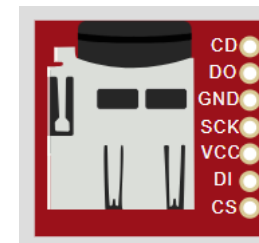
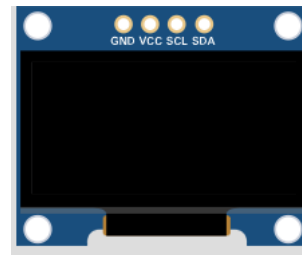
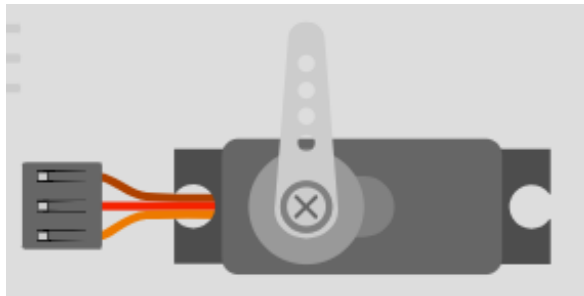
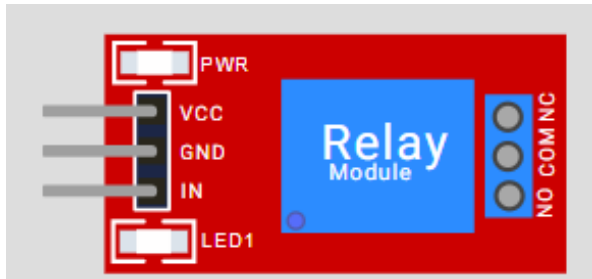


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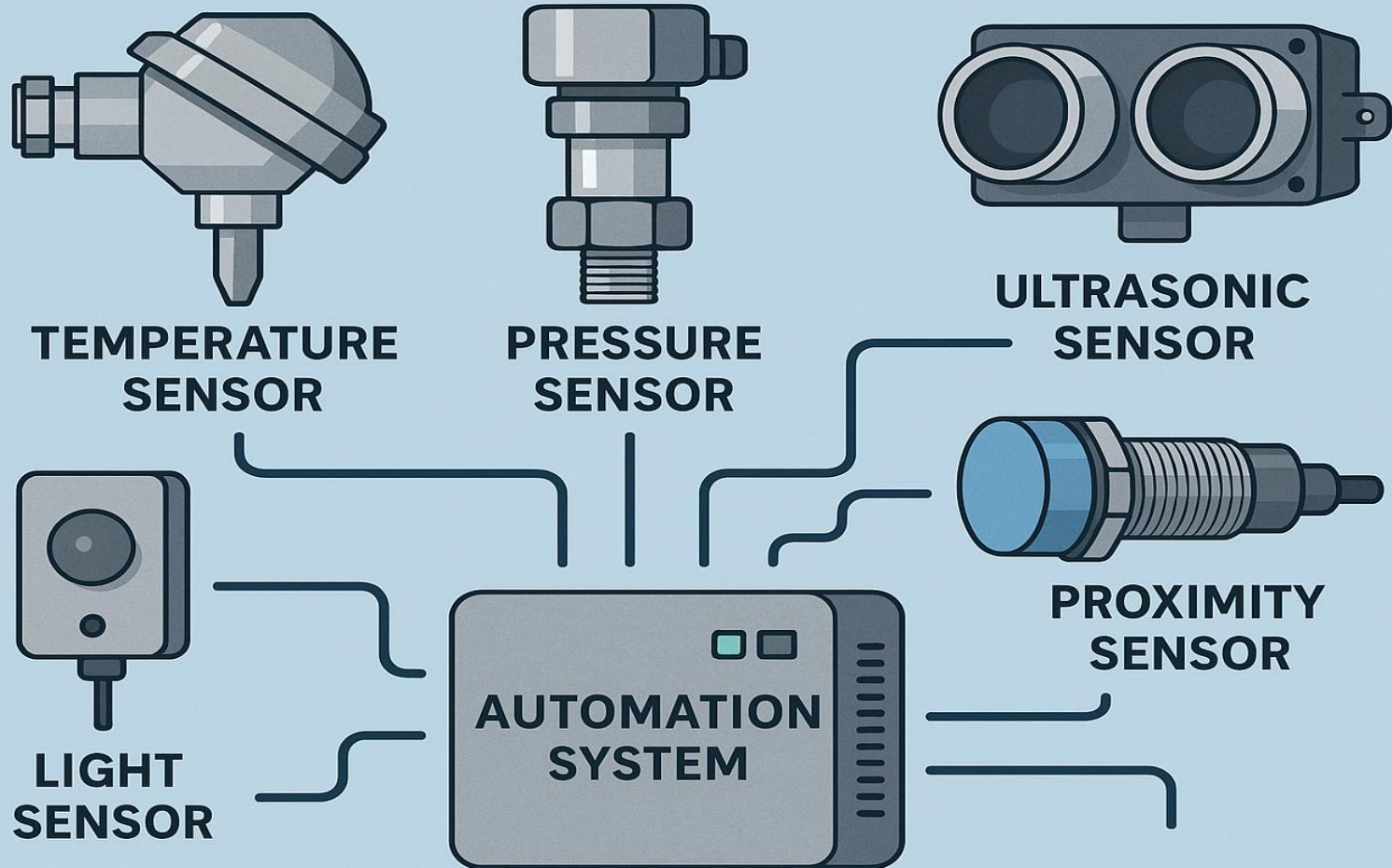
Sensors



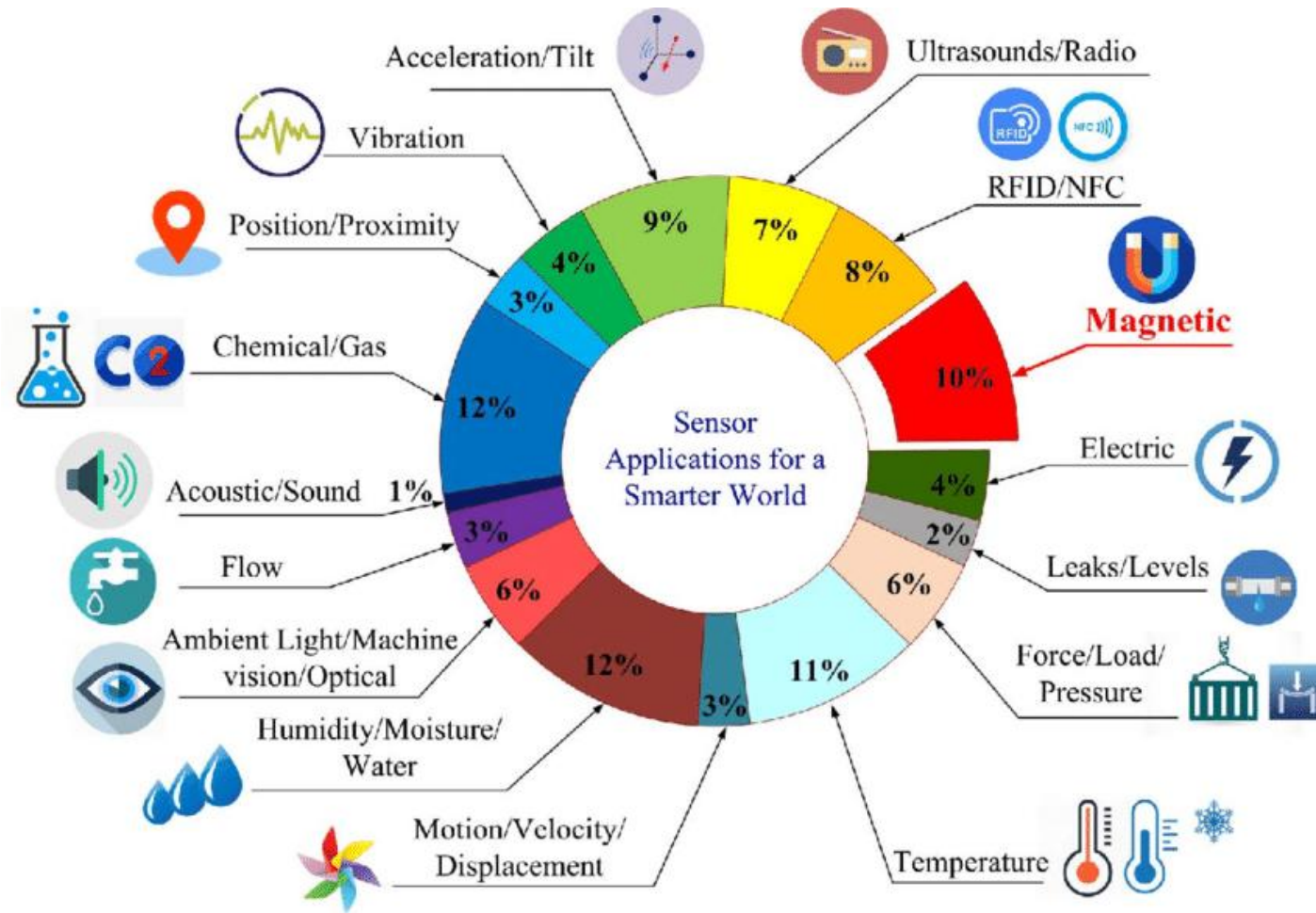
Actuators



1. Sensors

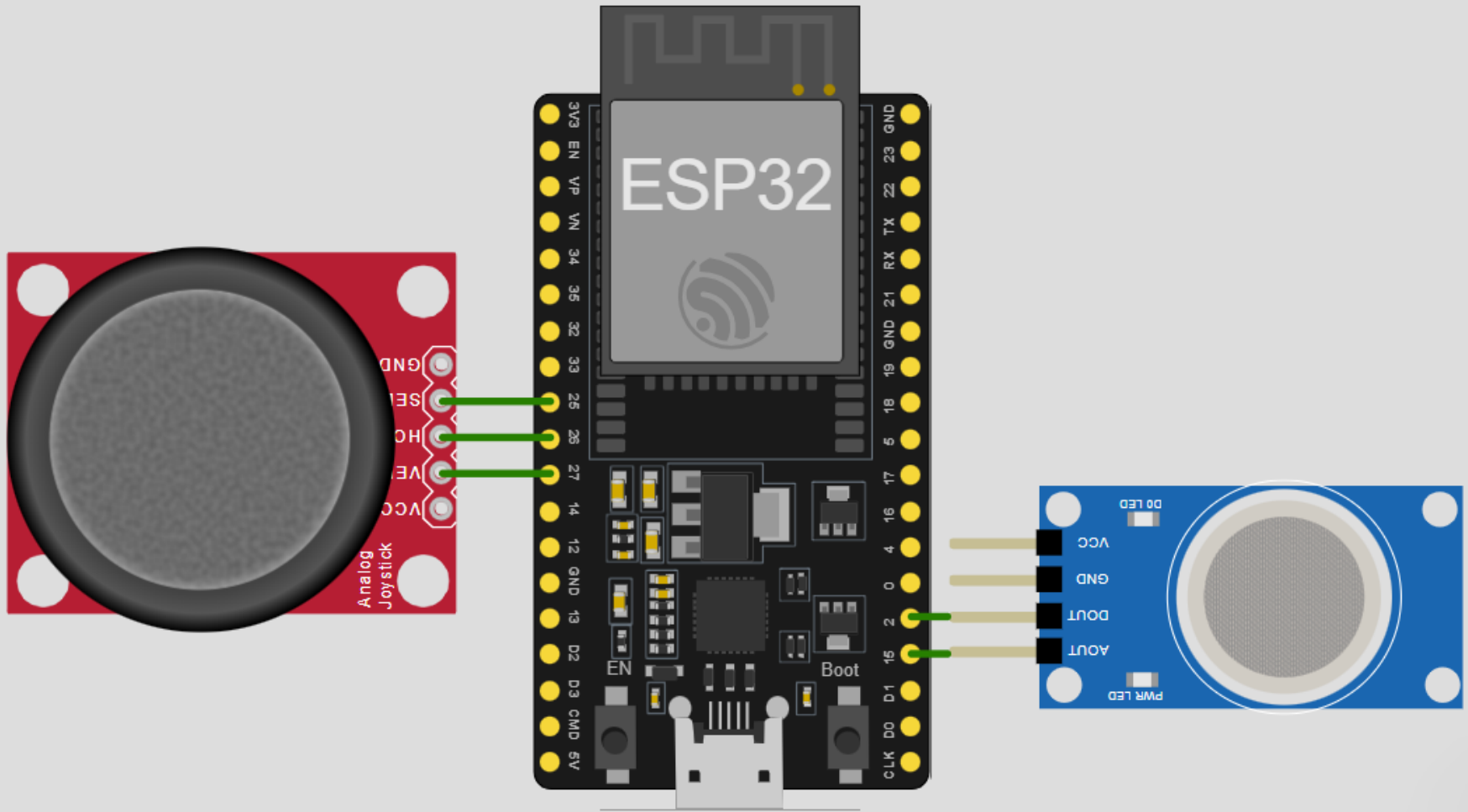


1. Sensors

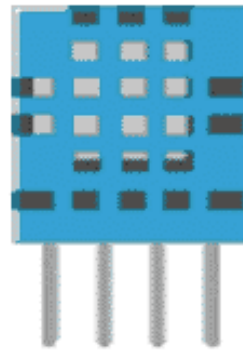


X. Liu *et al.*, "Overview of Spintronic Sensors With Internet of Things for Smart Living," in *IEEE Transactions on Magnetics*, vol. 55, no. 11, pp. 1-22, Nov. 2019.

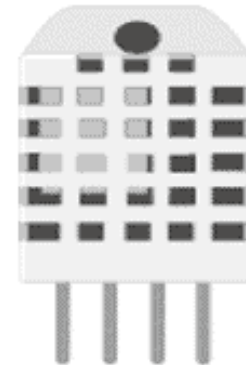
1. Sensors: Temperature, Humidity



1. Sensors



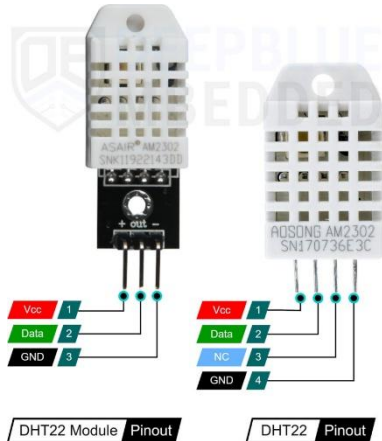
DHT11



DHT22

	DHT11	DHT22
Operating Voltage	3 to 5V	3 to 5V
Max Operating Current	2.5mA max	2.5mA max
Temperature Range	0-50°C / $\pm 2^{\circ}\text{C}$	-40 to 80°C / $\pm 0.5^{\circ}\text{C}$
Humidity Range	20-80% / 5%	0-100% / 2-5%
Sampling Rate	1 Hz (reading every second)	0.5 Hz (reading every 2 seconds)
Advantage	low cost	More Accurate

1. Sensors: Temperature, Humidity

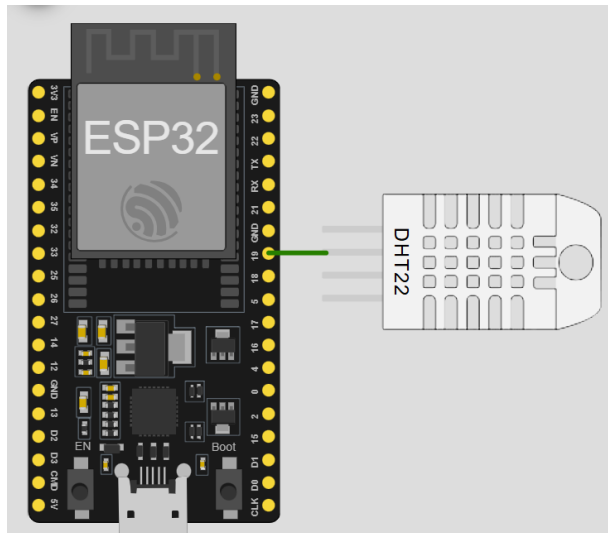


Libraries 114

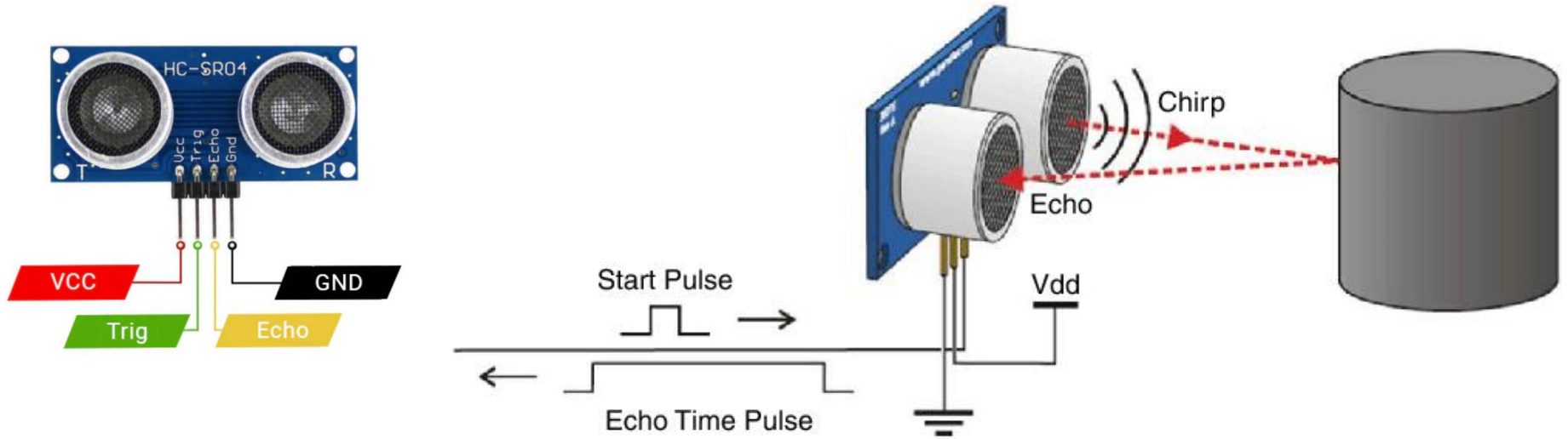
DHT sensor library by Adafruit

```
15 lib_deps =  
16   adafruit/DHT sensor library @ ^1.4.6
```

```
1  #include <Arduino.h>  
2  #include "DHT.h"  
3  DHT dht(2,DHT22);  
4  void setup()  
5  {  
6      dht.begin();  
7  }  
8  void loop()  
9  {  
10     float h = dht.readHumidity();  
11     float t = dht.readTemperature();  
12     float f = dht.readTemperature(true);  
13 }
```



1. Sensors: Ultrasonic



Ultrasonic sensor

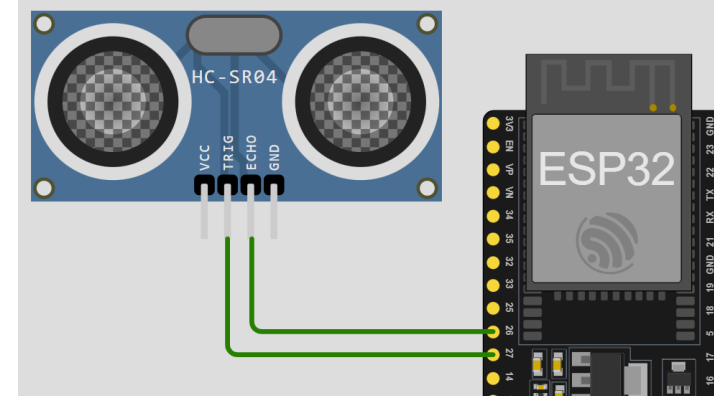
1. Sensors : Ultrasonic

Ultrasonic by Erick Simões

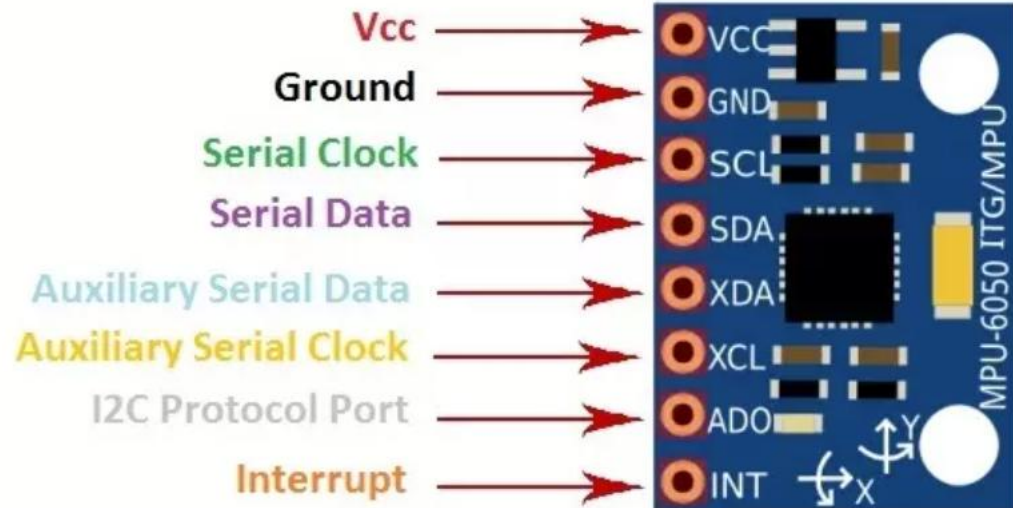
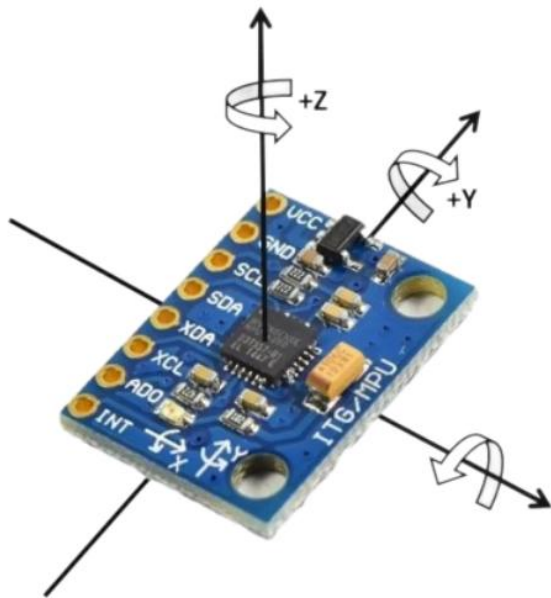
18
19

```
lib_deps =  
ericksimoes/Ultrasonic @ ^3.0.0
```

```
1  #include <Arduino.h>  
2  #include <Ultrasonic.h>  
3  Ultrasonic ultrasonic(12, 13);  
4  int distance;  
5  void setup()  
6  {  
7  }  
8  void loop()  
9  {  
10     distance = ultrasonic.read();  
11 }
```

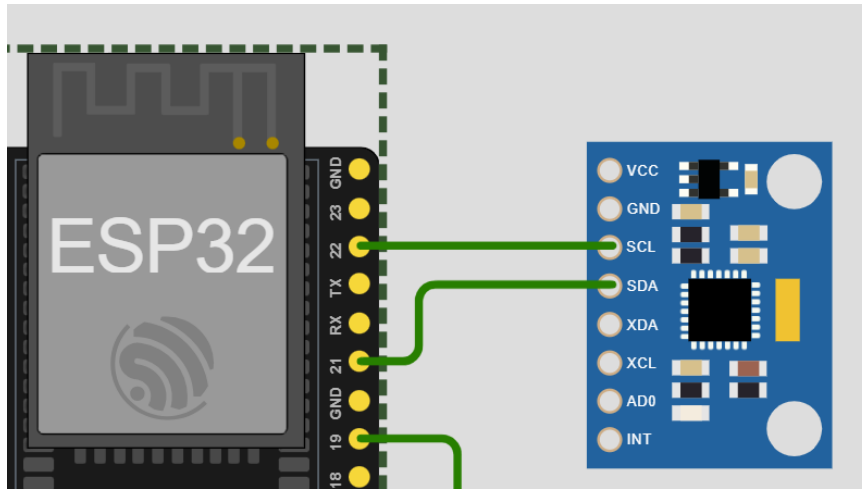


1. Sensors : Accelerator sensor



MPU 6050

1. Sensors: Accelerator sensor



Libraries 55

[Adafruit MPU6050](#) by Adafruit

```
15 lib_deps =  
16     # RECOMMENDED  
17     # Accept new functionality in a backwards compatible manner  
18     adafruit/Adafruit MPU6050 @ ^2.2.6
```

```
1  #include <Adafruit_MPU6050.h>  
2  #include <Adafruit_Sensor.h>  
3  #include <Wire.h>  
4  Adafruit_MPU6050 mpu;
```

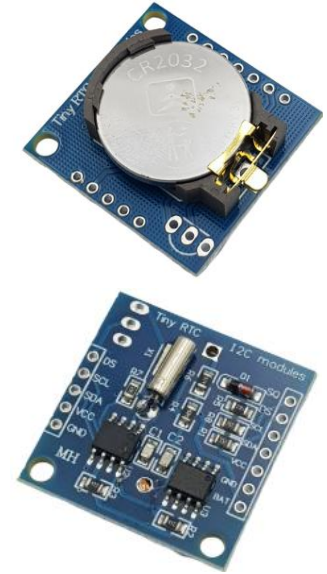
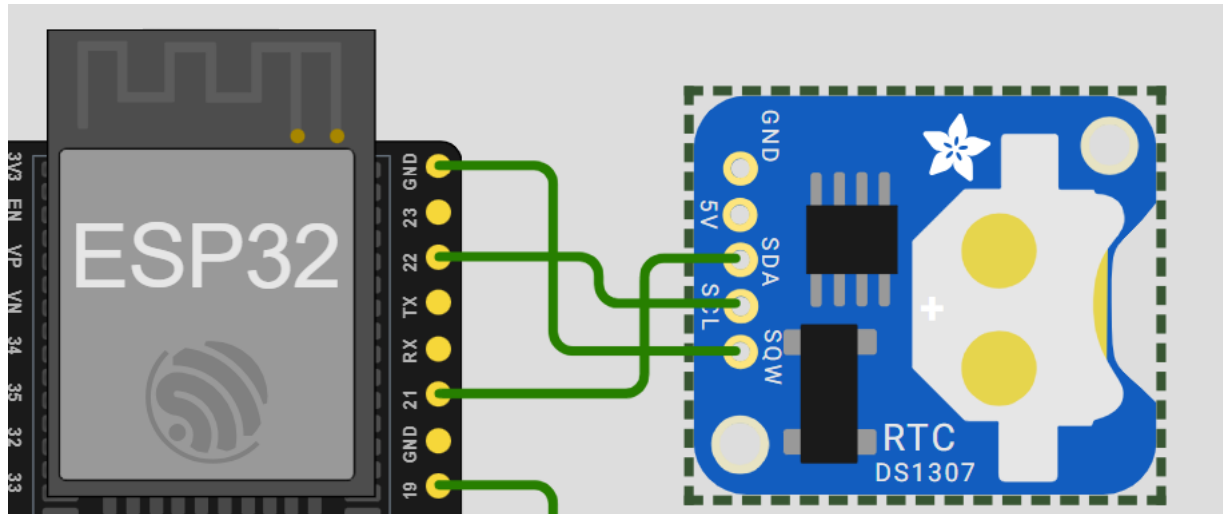
1. Sensors: Accelerator sensor

```
5  void setup(void)
6  {
7      Serial.begin(115200);
8      if (!mpu.begin())
9          while (1) ;
10     Serial.println("MPU6050 Found!");
11     //setupt motion detection
12     mpu.setHighPassFilter(MPU6050_HIGHPASS_0_63_HZ);
13     mpu.setMotionDetectionThreshold(1);
14     mpu.setMotionDetectionDuration(20);
15     mpu.setInterruptPinLatch(true);    // Keep it latched.  Will t
16     mpu.setInterruptPinPolarity(true);
17     mpu.setMotionInterrupt(true);
18     Serial.println("");
19     delay(100);
20 }
```


1. Sensors: Accelerator sensor

```
21  void loop()  
22  {  
23      sensors_event_t a, g, temp;  
24      mpu.getEvent(&a, &g, &temp);  
25      Serial.print(a.acceleration.x);  
26      Serial.print(a.acceleration.y);  
27      Serial.print(a.acceleration.z);  
28      Serial.print(g.gyro.x);  
29      Serial.print(g.gyro.y);  
30      Serial.print(g.gyro.z);  
31  }
```

1. Sensors: RTC module



DS1307
Real Time Module

Libraries 65

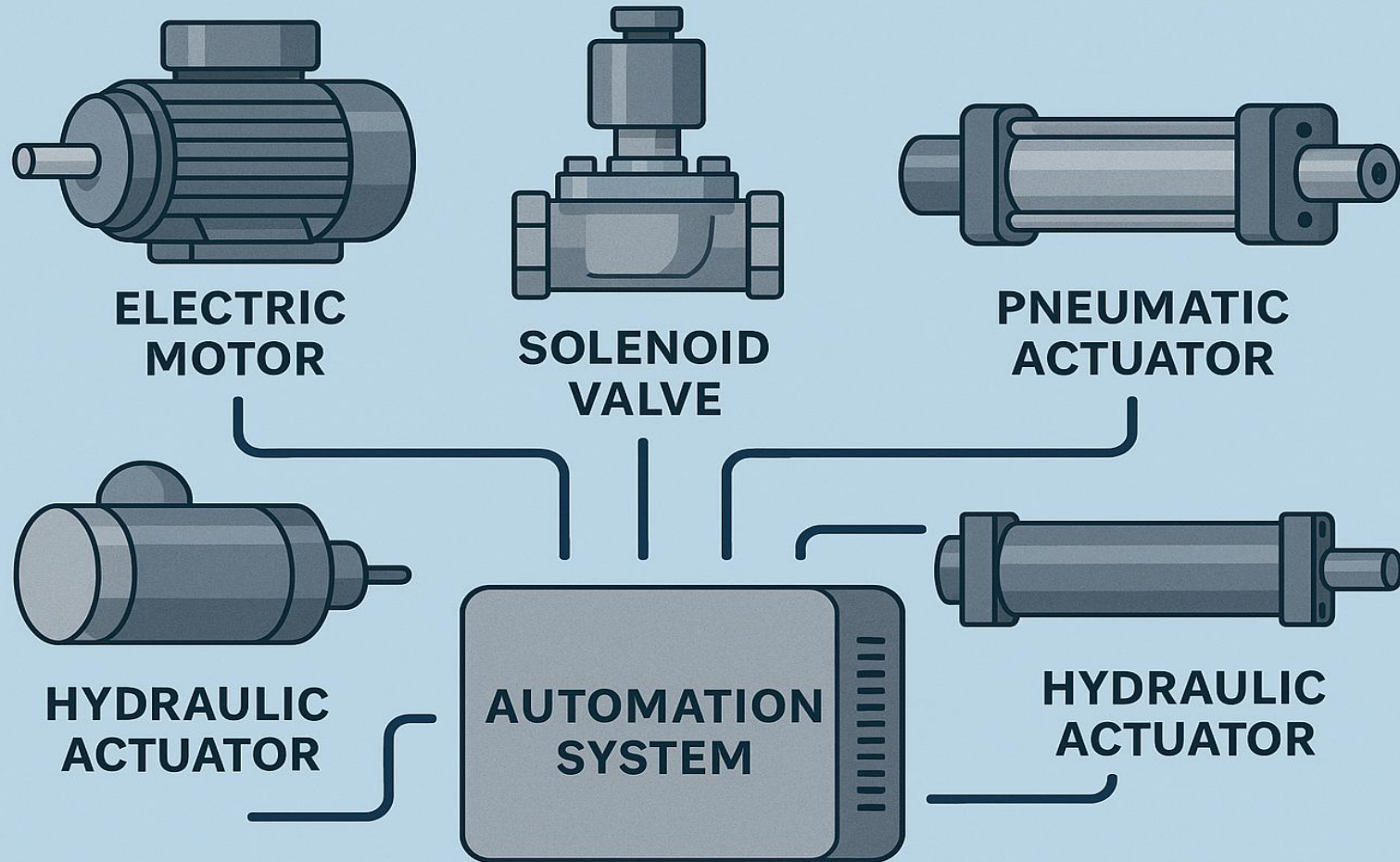
[RTClib](#) by Adafruit 83,705 15 Arduino

```
lib_deps =  
  # RECOMMENDED  
  # Accept new functionality in a backwards compatible manner and patches  
  adafruit/RTClib @ ^2.1.4
```

1. Sensors : RTC module

```
1  #include <Arduino.h>
2  #include "RTClib.h"
3  RTC_DS1307 rtc;
4  void setup()
5  {
6      if (!rtc.begin())
7      {
8          Serial.println("Couldn't find RTC");
9          while (1);
10     }
11     rtc.adjust(DateTime(2014, 1, 21, 3, 34, 16));
12 }
13 void loop()
14 {
15     DateTime now = rtc.now();
16     int y = now.year();
17     int m = now.month();
18     int d = now.day();
19     int h = now.hour();
20     int min = now.minute();
21     int s = now.second();
22 }
```

2. Actuators

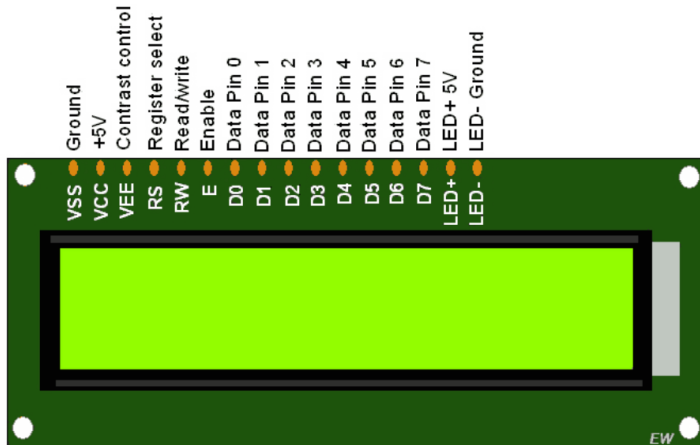


2. Actuators

Types of Actuators



2. Actuators: LCD 16x2



Module LCD 16x2

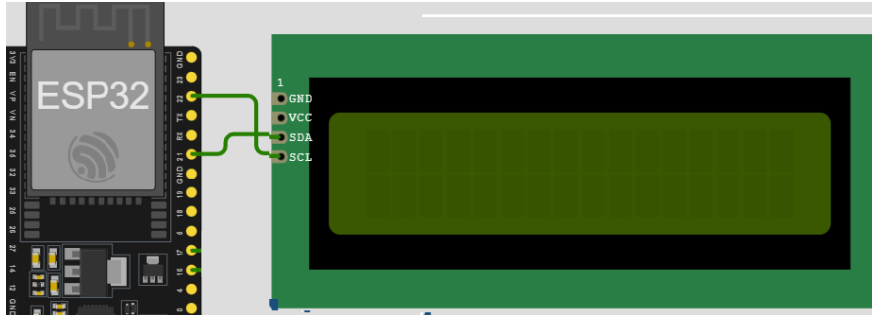


Grove LCD



Module LCD with IC
HD44780

2. Actuators : LCD 16x2

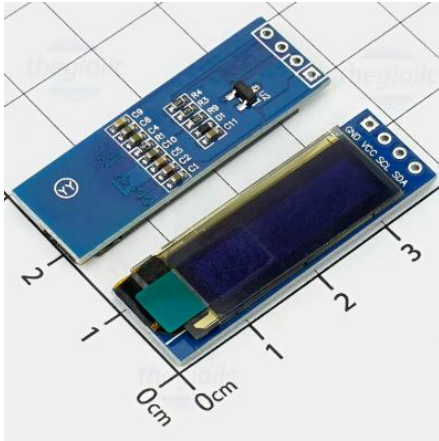


LiquidCrystal_I2C by Frank de Brabander

```
15 lib_deps =  
16 | marcoschwartz/LiquidCrystal_I2C @ ^1.1.4
```

```
1  #include <Arduino.h>  
2  #include <Wire.h>  
3  #include <LiquidCrystal_I2C.h>  
4  LiquidCrystal_I2C lcd(0x27,16,2);  
5  void setup()  
6  {  
7      lcd.init();  
8      lcd.backlight();  
9  }  
10 void loop()  
11 {  
12     lcd.clear();  
13     lcd.setCursor(0,0);  
14     lcd.print("Hello LCD");  
15 }
```

2. Actuators : OLED LCD



Oled LCD 128x32



Oled LCD 128x64

Libraries 116

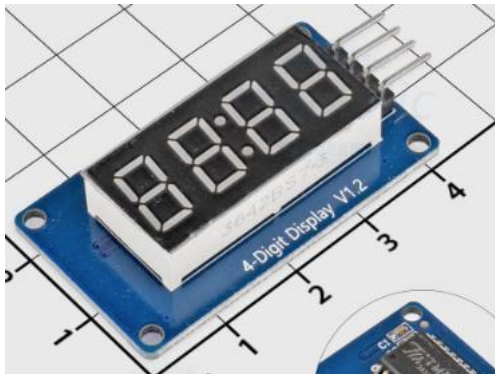
[Adafruit SSD1306](#) by Adafruit

```
15  lib_deps =
16  |  adafruit/Adafruit SSD1306 @ ^2.5.13
```

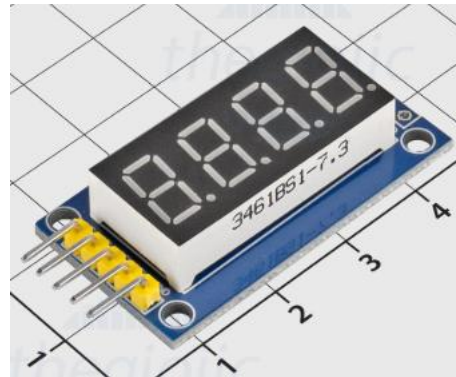

2. Actuators : Oled LCD

```
1  #include <Arduino.h>
2  #include <Wire.h>
3  #include <Adafruit_SSD1306.h>
4
5  #define SCREEN_WIDTH 128    // OLED display width, in pixels
6  #define SCREEN_HEIGHT 64    // OLED display height, in pixels
7  #define OLED_RESET      -1  // Reset pin # (or -1 if sharing Arduino reset pin)
8  #define SCREEN_ADDRESS 0x3C ///< See datasheet for Address; 0x3D for 128x64, 0x3C for
9  Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
10
11 void setup()
12 {
13   if (!display.begin(SSD1306_SWITCHCAPVCC, SCREEN_ADDRESS))
14     for (;;); // Don't proceed, loop forever
15   display.clearDisplay();
16   display.setTextSize(1);
17   display.setTextColor(SSD1306_WHITE);
18   display.setCursor(0,0);
19   display.println("Connecting to WiFi...");
20   display.display();
21 }
22 void loop()
23 {
24 }
```

2. Actuators : 7-segment module



7-segment
TM1367 Module



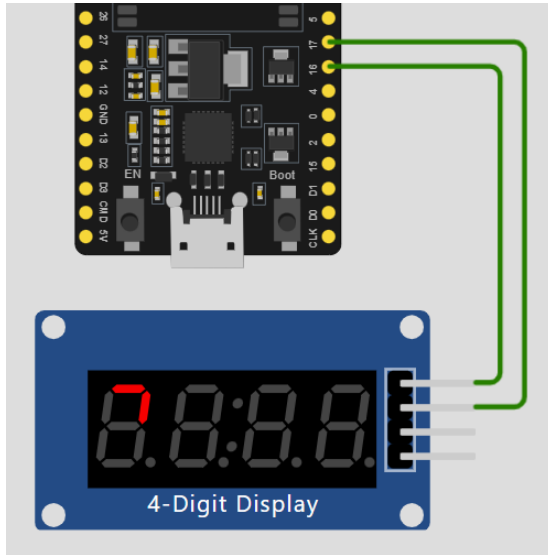
7-segment
74HC595 Module

Libraries 26

TM1637 by Avishay

```
15 lib_deps =
16 | smougenot/TM1637 @ 0.0.0-alpha+sha.9486982048
```

2. Actuators : 7-segment module

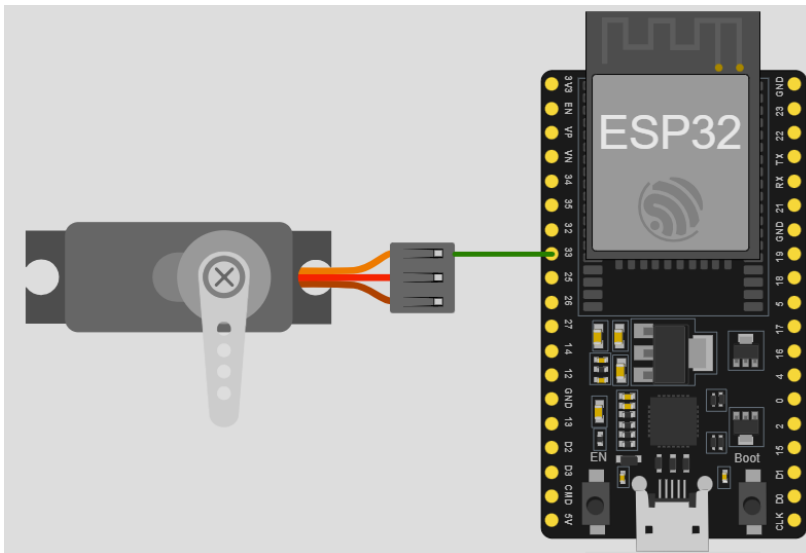


```
1  #include <Arduino.h>
2  #include <TM1637Display.h>
3  #define CLK 16
4  #define DIO 17
5  TM1637Display dis(CLK, DIO);
6  void setup()
7  {
8      uint8_t data[] = { 0xff, 0xff, 0xff, 0xff };
9      dis.setBrightness(0x0f);
10     dis.setSegments(data);
11     delay(3000);
12     //Selectively set different digits
13     data[0] = dis.encodeDigit(0);
14     data[1] = dis.encodeDigit(1);
15     data[2] = dis.encodeDigit(2);
16     data[3] = dis.encodeDigit(3);
17     dis.setSegments(data);
18 }
19 void loop()
20 {
21 }
```

2. Actuators : RC Servo



RC-Servo

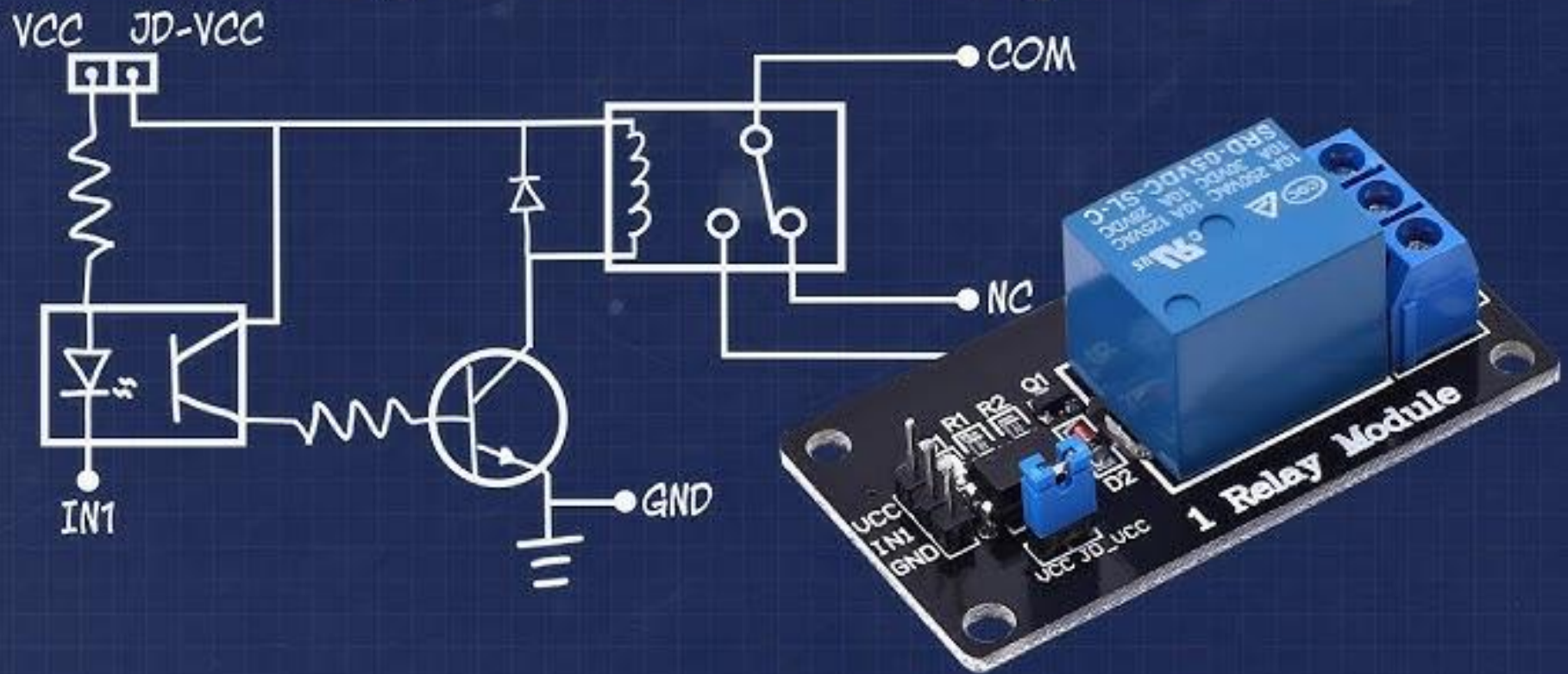


ServoESP32 by Jaroslav Paral

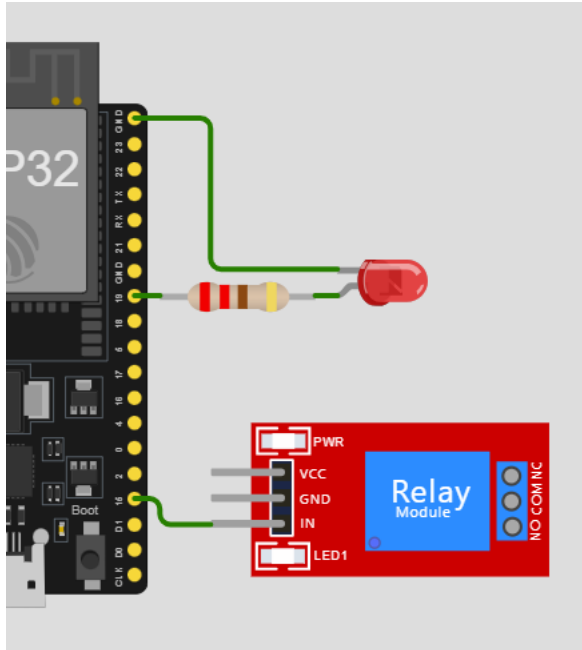
```
15 lib_deps =  
16 # RECOMMENDED  
17 # Accept new functionality in a backwards compatible manner and  
18 roboticsbrno/ServoESP32 @ ^1.1.1
```

2. Actuators : Relay module

Relay Modules Explained



2. Actuators : Relay module



```
1  #include <Arduino.h>
2  byte counter;
3  void setup()
4  {
5      pinMode(15,OUTPUT);
6      pinMode(19,OUTPUT);
7  }
8  void loop()
9  {
10     counter+=20;
11     digitalWrite(15,HIGH); // turn the LED on
12     delay(100);           // wait for a second
13     digitalWrite(15,LOW); // turn the LED off
14     delay(100);
15     analogWrite(19,counter); // wait for
16 }
```

Thank for attention